

Unit-II.

FOURIER TRANSFORMS

Fourier integral theorem (without proof) - Fourier transform pair
- Sine and Cosine transforms - Properties - Transforms of simple
functions - Convolution theorem - Parseval's identity.

2.0 INTRODUCTION

Integral transforms are used in the solution of partial differential equations. The choice of a particular transform to be used for the solution of a differential equation depends upon the nature of the boundary conditions of the equation and the facility with which the transform $F(s)$ can be inverted to give $f(x)$.

INTEGRAL TRANSFORMS

The integral transform $F(s)$ of a function $f(x)$ is defined as

$$I[f(x)] = F(s) = \int_a^b f(x) k(s, x) dx$$

For example,

(i) Laplace Transform $k(s, x) = e^{-sx}$

$$L[f(x)] = F(s) = \int_0^{\infty} e^{-sx} f(x) dx$$

(ii) Fourier complex transform $k(s, x) = e^{-isx}$

$$F[f(x)] = F(s) = \int_{-\infty}^{\infty} f(x) e^{-isx} dx$$

(iii) Fourier sine transform $k(s, x) = \sin sx$

$$F_s[f(x)] = F(s) = \int_0^{\infty} f(x) \sin sx dx$$

(iv) Fourier cosine transform $k(s, x) = \cos sx$

$$F_c[f(x)] = F(s) = \int_0^{\infty} f(x) \cos sx \, dx$$

(v) Hankel transform $k(s, s) = x J_n(sx)$

$$H[f(x)] = F(s) = \int_0^{\infty} x J_n(sx) f(x) \, dx$$

(vi) Mellin transform $k(s, x) = x^{s-1}$

$$M[f(x)] = F(s) = \int_0^{\infty} f(x) x^{s-1} \, dx$$

Note : Integral transforms are used in solving the partial differential equation with boundary conditions.

LIST OF FORMULAE

Fourier Integrals and Transforms :

1. Fourier Integral for $f(x)$ is

$$f(x) = \frac{1}{\pi} \int_0^{\infty} \int_{-\infty}^{\infty} f(t) \cos u(t-x) \, du \, dt$$

2. Fourier sine Integral for $f(x)$ is

$$f(x) = \frac{2}{\pi} \int_0^{\infty} \int_0^{\infty} f(t) \sin ut \sin ux \, du \, dt$$

3. Fourier cosine Integral for $f(x)$ is

$$f(x) = \frac{2}{\pi} \int_0^{\infty} \int_0^{\infty} f(t) \cos ut \cos ux \, du \, dt$$

4. Fourier Transform of $f(x)$ is $F(s) = \int_{-\infty}^{\infty} f(x) e^{isx} \, dx$

Inverse Fourier Transform of $F(s)$ is $f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(s) e^{-isx} \, ds$

FOURIER TRANSFORM

5. Fourier sine transform of $f(x)$ is $F_s(s) = \int_0^{\infty} f(x) \sin sx \, dx$

To get back $f(x)$, we have $f(x) = \frac{2}{\pi} \int_0^{\infty} F_s(s) \sin sx \, ds$

6. Fourier cosine Transform of $f(x)$ is $F_c(s) = \int_0^{\infty} f(x) \cos sx \, dx$

To get back $f(x)$, we get $f(x) = \frac{2}{\pi} \int_0^{\infty} F_c(s) \cos sx \, ds$

2.1 FOURIER INTEGRAL THEOREM

2.1.a Fourier integral theorem (without proof)

If $f(x)$ is piece-wise continuously differentiable and absolutely integrable in $(-\infty, \infty)$, then

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(t) e^{is(x-t)} \, dt \, ds$$

or equivalently

$$f(x) = \frac{1}{\pi} \int_0^{\infty} \int_{-\infty}^{\infty} f(t) \cos \lambda(t-x) \, dt \, d\lambda$$

This is known as Fourier integral theorem or Fourier integral formula.

(OR)

Let us assume the following conditions on a function $f(x)$

1. $f(x)$ is piece-wise continuous in any finite interval (a, b)
2. $\int_{-\infty}^{\infty} |f(x)| \, dx$ is convergent.

Then the Fourier integral theorem states that

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(t) \cos \lambda(t-x) \, dt \, d\lambda$$

The double integral in the right hand side is known as a Fourier integral expansion of $f(x)$

(OR)

If $f(x)$ is a function defined in $(-l, l)$ satisfying Dirichlet's conditions, then

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(t) \cos \lambda (t - x) dt d\lambda$$

The double integral in the right hand side is known as Fourier integral theorem or Fourier integral formula.

Note : We assumed the following conditions on $f(x)$

- (i) $f(x)$ is defined as single-valued except at finite points in $(-l, l)$
- (ii) $f(x)$ is periodic outside $(-l, l)$ with period $2l$.
- (iii) $f(x)$ and $f'(x)$ are sectionally continuous in $(-l, l)$

(iv) $\int_{-\infty}^{\infty} |f(x)| dx$ converges

i.e., $f(x)$ is absolutely integrable in $(-\infty, \infty)$

I. Problems based on Fourier integral theorem :

Example 2.1.a(1) : Show that $f(x) = 1, 0 < x < \infty$ cannot be represented by a Fourier integral.

Solution : $\int_0^{\infty} |f(x)| dx = \int_0^{\infty} 1 dx = [x]_0^{\infty} = \infty - 0 = \infty$

i.e., $\int_0^{\infty} |f(x)| dx$ is not convergent.

Hence $f(x) = 1$ cannot be represented by a Fourier integral.

Example 2.1.a(2) : If $f(x) = \begin{cases} \frac{\pi}{2}, & |x| < 1 \\ 0, & |x| > 1 \end{cases}$,

show that $f(x) = \int_0^{\infty} \frac{\cos \lambda x \sin \lambda}{\lambda} d\lambda$

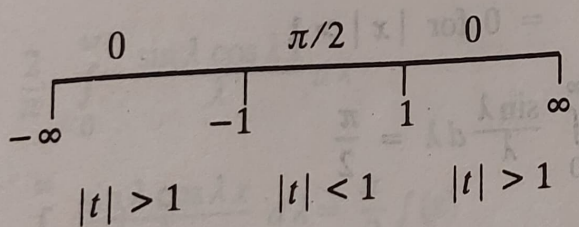
Hence show that $\int_0^{\infty} \frac{\sin \lambda}{\lambda} d\lambda = \frac{\pi}{2}$

Solution : We know that the Fourier integral formula for $f(x)$ is

$$f(x) = \frac{1}{\pi} \int_0^{\infty} \int_{-\infty}^{\infty} f(t) \cos \lambda (t-x) dt d\lambda \quad \dots (1)$$

Here $f(t) = \frac{\pi}{2}$ for $|t| < 1$ is $-1 < t < 1$

$f(t) = 0$ for $|t| > 1$ i.e., $-\infty < t < -1$ and $1 < t < \infty$



$$(1) \Rightarrow f(x) = \frac{1}{\pi} \int_0^{\infty} \int_{-1}^1 \frac{\pi}{2} \cos \lambda (t-x) dt d\lambda$$

$$= \frac{1}{\pi} \frac{\pi}{2} \int_0^{\infty} \int_{-1}^1 \cos \lambda (t-x) dt d\lambda$$

$$= \frac{1}{2} \int_0^{\infty} \left[\frac{\sin \lambda (t-x)}{\lambda} \right]_{-1}^1 d\lambda$$

$$= \frac{1}{2} \int_0^{\infty} \left[\frac{\sin \lambda (1-x)}{\lambda} - \frac{\sin \lambda (-1-x)}{\lambda} \right] d\lambda$$

$$= \frac{1}{2} \int_0^{\infty} \left[\frac{\sin \lambda (1-x)}{\lambda} + \frac{\sin \lambda (1+x)}{\lambda} \right] d\lambda$$

$$\begin{aligned}
 &= \frac{1}{2} \int_0^{\infty} \frac{\sin \lambda (1-x) + \sin \lambda (1+x)}{\lambda} d\lambda \\
 &= \frac{1}{2} \int_0^{\infty} \frac{\sin (\lambda - \lambda x) + \sin (\lambda + \lambda x)}{\lambda} d\lambda \\
 &= \frac{1}{2} \int_0^{\infty} \frac{2 \sin \lambda \cos \lambda x}{\lambda} d\lambda \quad [\because \sin (A+B) + \sin (A-B) \\
 &\qquad\qquad\qquad = 2 \sin A \cos B] \\
 &= \int_0^{\infty} \frac{\sin \lambda \cos \lambda x}{\lambda} d\lambda
 \end{aligned}$$

$$\text{i.e., } \int_0^{\infty} \frac{\sin \lambda \cos \lambda x}{\lambda} d\lambda = f(x)$$

$$= \frac{\pi}{2} \text{ for } |x| < 1$$

$$= 0 \text{ for } |x| > 1$$

$$\text{Putting } x = 0 \text{ we get } \int_0^{\infty} \frac{\sin \lambda}{\lambda} d\lambda = \frac{\pi}{2}$$

Example 2.1.a(3) : Express the function $f(x) = \begin{cases} 1 & \text{for } |x| \leq 1 \\ 0 & \text{for } |x| > 1 \end{cases}$

as a Fourier integral. Hence evaluate $\int_0^{\infty} \frac{\sin \lambda \cos \lambda x}{\lambda} d\lambda$ and find

the value of $\int_0^{\infty} \frac{\sin \lambda}{\lambda} d\lambda$.

[A.U. April, 2001]

Solution : We know that the Fourier integral formula for $f(x)$ is

$$f(x) = \frac{1}{\pi} \int_0^{\infty} \int_{-\infty}^{\infty} f(t) \cos \lambda (t-x) dt d\lambda \quad \dots (1)$$

Here $f(t) = 1$ for $|t| \leq 1$ i.e., $-1 \leq t \leq 1$

$f(t) = 0$ for $|t| > 1$ i.e., $-\infty < t < -1$ and $1 < t < \infty$

$$\begin{aligned}
 (1) \Rightarrow f(x) &= \frac{1}{\pi} \int_0^{\infty} \int_{-1}^1 \cos \lambda (t-x) dt d\lambda \\
 &= \frac{1}{\pi} \int_0^{\infty} \left[\frac{\sin \lambda (t-x)}{\lambda} \right]_{-1}^1 d\lambda \\
 &= \frac{1}{\pi} \int_0^{\infty} \left[\frac{\sin \lambda (1-x)}{\lambda} - \frac{\sin \lambda (-1-x)}{\lambda} \right] d\lambda \\
 &= \frac{1}{\pi} \int_0^{\infty} \left[\frac{\sin \lambda (1-x)}{\lambda} + \frac{\sin \lambda (1+x)}{\lambda} \right] d\lambda \quad [\because \sin(-\theta) = -\sin \theta] \\
 &= \frac{1}{\pi} \int_0^{\infty} \frac{\sin(\lambda - \lambda x) + \sin(\lambda + \lambda x)}{\lambda} d\lambda \\
 &= \frac{1}{\pi} \int_0^{\infty} \frac{2 \sin \lambda \cos \lambda x}{\lambda} d\lambda \quad [\because \sin(A+B) + \sin(A-B) = 2 \sin A \cos B] \\
 &= \frac{2}{\pi} \int_0^{\infty} \frac{\sin \lambda \cos \lambda x}{\lambda} d\lambda \\
 \text{i.e., } \int_0^{\infty} \frac{\sin \lambda \cos \lambda x}{\lambda} d\lambda &= \frac{\pi}{2} f(x) \\
 &= \frac{\pi}{2} \text{ for } |x| \leq 1 \\
 &= 0 \text{ for } |x| > 1
 \end{aligned}$$

Putting $x = 0$ we get $\int_0^{\infty} \frac{\sin \lambda}{\lambda} d\lambda = \frac{\pi}{2}$

Example 2.1.a(4) : Find the Fourier integral of the function

$$f(t) = \begin{cases} e^{at}, & t < 0 \\ e^{-at}, & t > 0 \end{cases}$$

Solution : We know that the Fourier integral formula for $f(x)$ is

$$f(x) = \frac{1}{\pi} \int_0^{\infty} \int_{-\infty}^{\infty} f(t) \cos \lambda (t-x) dt d\lambda \quad \dots (1)$$

Here $f(t) = e^{at}$ for $t < 0$ i.e., $-\infty < t < 0$

$f(t) = e^{-at}$ for $t > 0$ i.e., $0 < t < \infty$

$$(1) \Rightarrow f(x) = \frac{1}{\pi} \int_0^{\infty} \left[\int_{-\infty}^0 e^{at} \cos \lambda(t-x) dt + \int_0^{\infty} e^{-at} \cos \lambda(t-x) dt \right] d\lambda \quad \dots (2)$$

(i) To find $\int_{-\infty}^0 e^{at} \cos \lambda(t-x) dt$

$$= \int_{-\infty}^0 e^{at} \cos(\lambda t - \lambda x) dt$$

$$= \int_{-\infty}^0 e^{at} [\cos \lambda t \cos \lambda x + \sin \lambda t \sin \lambda x] dt$$

$$= \int_{-\infty}^0 e^{at} \cos \lambda t \cos \lambda x dt + \int_{-\infty}^0 e^{at} \sin \lambda t \sin \lambda x dt$$

$$= \cos \lambda x \int_{-\infty}^0 e^{at} \cos \lambda t dt + \sin \lambda x \int_{-\infty}^0 e^{at} \sin \lambda t dt$$

$$= \cos \lambda x \left[\frac{e^{at}}{a^2 + \lambda^2} [a \cos \lambda t + \lambda \sin \lambda t] \right]_{-\infty}^0$$

$$+ \sin \lambda x \left[\frac{e^{at}}{a^2 + \lambda^2} [a \sin \lambda t - \lambda \cos \lambda t] \right]_{-\infty}^0$$

$$= \cos \lambda x \left[\frac{a}{a^2 + \lambda^2} - 0 \right] + \sin \lambda x \left[\frac{-\lambda}{a^2 + \lambda^2} - 0 \right]$$

$$= \frac{a \cos \lambda x - \lambda \sin \lambda x}{a^2 + \lambda^2}$$

(ii) To find $\int_0^{\infty} e^{-at} \cos \lambda (t-x) dt$

$$= \int_0^{\infty} e^{-at} \cos (\lambda t - \lambda x) dt$$

$$= \int_0^{\infty} e^{-at} [\cos \lambda t \cos \lambda x + \sin \lambda t \sin \lambda x] dt$$

$$= \cos \lambda x \int_0^{\infty} e^{-at} \cos \lambda t dt + \sin \lambda x \int_0^{\infty} e^{-at} \sin \lambda t dt$$

$$= \cos \lambda x \left[\frac{a}{a^2 + \lambda^2} \right]$$

$$+ \sin \lambda x \left[\frac{\lambda}{a^2 + \lambda^2} \right]$$

$$\text{Formula : } \int_0^{\infty} e^{-ax} \cos bx dx = \frac{a}{a^2 + b^2};$$

$$\int_0^{\infty} e^{-ax} \sin bx dx = \frac{b}{a^2 + b^2}$$

$$= \frac{a \cos \lambda x + \lambda \sin \lambda x}{a^2 + \lambda^2}$$

Now (2) $\Rightarrow$

$$f(x) = \frac{1}{\pi} \int_0^{\infty} \left[\frac{a \cos \lambda x - \lambda \sin \lambda x}{a^2 + \lambda^2} + \frac{a \cos \lambda x + \lambda \sin \lambda x}{a^2 + \lambda^2} \right] d\lambda$$

$$= \frac{1}{\pi} \int_0^{\infty} \frac{a \cos \lambda x - \lambda \sin \lambda x + a \cos \lambda x + \lambda \sin \lambda x}{a^2 + \lambda^2} d\lambda$$

$$= \frac{1}{\pi} \int_0^{\infty} \frac{2a \cos \lambda x}{a^2 + \lambda^2} d\lambda = \frac{2a}{\pi} \int_0^{\infty} \frac{\cos \lambda x}{a^2 + \lambda^2} d\lambda$$

$$\text{i.e., } \int_0^{\infty} \frac{\cos \lambda x}{a^2 + \lambda^2} d\lambda = \frac{\pi}{2a} f(x)$$

$$= \frac{\pi}{2a} e^{ax}, \quad x < 0$$

$$= \frac{\pi}{2a} e^{-ax}, \quad x > 0$$

Example 2.1.a(5) : Find the Fourier integral of the function

$$f(x) = \begin{cases} 0, & x < 0 \\ 1/2, & x = 0 \\ e^{-x}, & x > 0 \end{cases} \text{ verify the representation directly at the point}$$

$$x = 0.$$

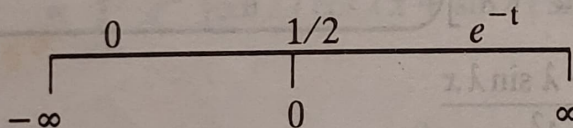
Solution : We know that the Fourier integral of $f(x)$ is

$$f(x) = \frac{1}{\pi} \int_0^{\infty} \int_{-\infty}^{\infty} f(t) \cos \lambda (t-x) dt d\lambda \quad \dots (1)$$

Here $f(t) = 0$ for $t < 0$ i.e., $-\infty < t < 0$

$$= \frac{1}{2} \text{ for } t = 0$$

$$= e^{-t} \text{ for } t > 0 \text{ i.e., } 0 < t < \infty$$



$$\begin{aligned} (1) \Rightarrow f(x) &= \frac{1}{\pi} \int_0^{\infty} \int_0^{\infty} e^{-t} \cos \lambda (t-x) dt d\lambda \\ &= \frac{1}{\pi} \int_0^{\infty} \int_0^{\infty} e^{-t} \cos (\lambda t - \lambda x) dt d\lambda \\ &= \frac{1}{\pi} \int_0^{\infty} \int_0^{\infty} e^{-t} [\cos \lambda t \cos \lambda x + \sin \lambda t \sin \lambda x] dt d\lambda \\ &= \frac{1}{\pi} \int_0^{\infty} \left[\cos \lambda x \int_0^{\infty} e^{-t} \cos \lambda t dt + \sin \lambda x \int_0^{\infty} e^{-t} \sin \lambda t dt \right] d\lambda \\ &= \frac{1}{\pi} \int_0^{\infty} \left[\cos \lambda x \left[\frac{1}{1+\lambda^2} \right] + \sin \lambda x \left[\frac{\lambda}{1+\lambda^2} \right] \right] d\lambda \end{aligned}$$

$$= \frac{1}{\pi} \int_0^{\infty} \frac{\cos \lambda x}{1 + \lambda^2}$$

$$+ \frac{\lambda \sin \lambda x}{1 + \lambda^2} d\lambda$$

$$= \frac{1}{\pi} \int_0^{\infty} \frac{\cos \lambda x + \lambda \sin \lambda x}{1 + \lambda^2} d\lambda$$

$$\text{i.e., } \int_0^{\infty} \frac{\cos \lambda x + \lambda \sin \lambda x}{1 + \lambda^2} d\lambda = \pi f(x)$$

$$= \pi \begin{cases} 0, & x < 0 \\ 1/2, & x = 0 \\ e^{-x}, & x > 0 \end{cases}$$

Putting $x = 0$ we get

$$\int_0^{\infty} \frac{1}{1 + \lambda^2} d\lambda = \pi f(0)$$

$$\text{i.e., } \pi f(0) = \int_0^{\infty} \frac{1}{1 + \lambda^2} d\lambda$$

$$\pi f(0) = \left[\tan^{-1} \lambda \right]_0^{\infty} = \left[\frac{\pi}{2} - 0 \right]$$

$$f(0) = \frac{1}{2}$$

The value of the given function at $x = 0$ is $\frac{1}{2}$.

Hence verified.

2.1.b Complex form of the Fourier integrals.

The Fourier integral formula for $f(x)$ is

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(t) \cos \lambda (t - x) dt d\lambda \quad \dots (1)$$

because $\cos \lambda (1-x)$ is an even function of λ .

Also since $\sin \lambda (t-x)$ is an odd function of λ .

$$\text{We have } \int_{-\infty}^{\infty} f(t) \sin \lambda (t-x) dt = 0$$

$$\text{i.e., } \int_{-\infty}^{\infty} f(t) i \sin \lambda (t-x) dt = 0$$

$$\begin{aligned} (1) \Rightarrow f(x) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(t) [\cos \lambda (t-x) + i \sin \lambda (t-x)] dt d\lambda \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(t) e^{i\lambda (t-x)} dt d\lambda \end{aligned}$$

which is the complex form of the Fourier integral.

FOURIER SINE AND COSINE INTEGRALS

$$(i) f(x) = \frac{2}{\pi} \int_0^{\infty} \sin ux du \int_0^{\infty} f(t) \sin ut dt$$

[Fourier sine Integral]

$$(ii) f(x) = \frac{2}{\pi} \int_0^{\infty} \cos ux du \int_0^{\infty} f(t) \cos ut dt$$

[Fourier cosine Integral]

Proof : We know that

$$\cos u(t-x) = \cos (ut - ux)$$

$$= \cos ut \cos ux + \sin ut \sin ux$$

By Fourier Integral Theorem

$$f(x) = \frac{1}{\pi} \int_0^{\infty} \int_0^{\infty} f(t) \cos u (t-x) dt du \quad \dots (1)$$

$$\begin{aligned}
 &= \frac{1}{\pi} \int_0^{\infty} \int_{-\infty}^{\infty} f(t) [\cos ut \cos ux + \sin ut \sin ux] du dt \\
 &= \frac{1}{\pi} \int_0^{\infty} \int_{-\infty}^{\infty} f(t) \cos ut \cos ux du dt + \frac{1}{\pi} \int_0^{\infty} \int_{-\infty}^{\infty} f(t) \sin ut \sin ux du dt \dots (2)
 \end{aligned}$$

Case (i) When $f(t)$ is odd

$$\therefore f(t) \cos ut \text{ is odd hence } \int_0^{\infty} \int_{-\infty}^{\infty} f(t) \cos ut \cos ux du dt = 0$$

$$[\text{For odd, } \int_{-a}^a f(x) dx = 0]$$

$$\text{For even, } \int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx]$$

From (2) we have

$$f(x) = \frac{2}{\pi} \int_0^{\infty} \sin ux du \int_0^{\infty} f(t) \sin ut dt \dots (3)$$

This is called Fourier sine integral

Case (ii) When $f(t)$ is even

$$\therefore f(t) \sin ut \text{ is odd, } \int_0^{\infty} \int_{-\infty}^{\infty} f(t) \sin ut \sin ux du dt = 0$$

$$\therefore f(t) \cos ut \text{ is even}$$

From (2) we have

$$f(x) = \frac{2}{\pi} \int_0^{\infty} \cos ux du \int_0^{\infty} f(t) \cos ut dt \dots (4)$$

This relation is known as Fourier cosine integral.

II. Problems based on Fourier cosine and Fourier sine integrals

Example 2.1.b(1) : Find Fourier cosine integral of the function

$$e^{-ax}. \text{ Hence deduce the value of the integral } \int_0^{\infty} \frac{\cos \lambda x}{1 + \lambda^2} d\lambda$$

Solution : We know that the Fourier cosine integral of the function $f(x)$ is given by

$$f(x) = \frac{2}{\pi} \int_0^{\infty} \cos \lambda x \int_0^{\infty} f(t) \cos \lambda t dt d\lambda \quad \dots (1)$$

Here $f(t) = e^{-at}$

$$\begin{aligned} (1) \Rightarrow f(x) &= \frac{2}{\pi} \int_0^{\infty} \cos \lambda x \int_0^{\infty} e^{-at} \cos \lambda t dt d\lambda \\ &= \frac{2}{\pi} \int_0^{\infty} \cos \lambda x \left[\frac{e^{-at}}{\lambda^2 + a^2} [-a \cos \lambda t + \lambda \sin \lambda t] \right]_0^{\infty} d\lambda \\ &= \frac{2}{\pi} \int_0^{\infty} \cos \lambda x \left[0 - \frac{1}{\lambda^2 + a^2} (-a) \right] d\lambda \\ &= \frac{2}{\pi} \int_0^{\infty} \frac{a \cos \lambda x}{\lambda^2 + a^2} d\lambda \\ &= \frac{2a}{\pi} \int_0^{\infty} \frac{\cos \lambda x}{\lambda^2 + a^2} d\lambda \end{aligned}$$

$$\text{i.e., } \int_0^{\infty} \frac{\cos \lambda x}{\lambda^2 + a^2} d\lambda = \frac{\pi}{2a} f(x) = \frac{\pi}{2a} e^{-ax}$$

Putting $a = 1$ we get

$$\int_0^{\infty} \frac{\cos \lambda x}{\lambda^2 + 1} d\lambda = \frac{\pi}{2} e^{-x}$$

Example 2.1.b(2) : Express $f(x) = \begin{cases} 1, & 0 \leq x \leq \pi \\ 0, & x > \pi \end{cases}$ as a Fourier

sine integral and hence evaluate $\int_0^{\infty} \frac{1 - \cos \pi \lambda}{\lambda} \sin(x\lambda) d\lambda$

Solution : W.K.T, the Fourier sine integral of the function $f(x)$ is given by

$$f(x) = \frac{2}{\pi} \int_0^{\infty} \sin \lambda x \int_0^{\infty} f(t) \sin \lambda t dt d\lambda \quad \dots (1)$$

$$= \frac{2}{\pi} \int_0^{\infty} \sin \lambda x \int_0^{\pi} \sin \lambda t dt d\lambda$$

$$\text{Since } f(t) = \begin{cases} 1, & 0 \leq t \leq \pi \\ 0, & t > \pi \end{cases}$$

$$= \frac{2}{\pi} \int_0^{\infty} \sin \lambda x \left[\frac{-\cos \lambda t}{\lambda} \right]_0^{\pi} d\lambda$$

$$= \frac{2}{\pi} \int_0^{\infty} \sin \lambda x \left[\frac{-\cos \lambda \pi}{\lambda} + \frac{1}{\lambda} \right] d\lambda$$

$$= \frac{2}{\pi} \int_0^{\infty} \sin \lambda x \frac{1 - \cos \lambda \pi}{\lambda} d\lambda$$

$$\therefore \int_0^{\infty} \frac{1 - \cos \lambda \pi}{\lambda} \sin \lambda x d\lambda = \frac{\pi}{2} f(x) = \begin{cases} \pi/2, & 0 < x < \pi \\ 0, & x > \pi \end{cases}$$

At $x = \pi$, which is a point of discontinuity of $f(x)$, the value of the above integral.

$$= \frac{\pi}{2} \left[\frac{f(\pi - 0) + f(\pi + 0)}{2} \right]$$

$$= \frac{\pi}{2} \left[\frac{1 + 0}{2} \right] = \frac{\pi}{4}$$

Example 2.1.b(3) : Using the Fourier integral representation show

$$\text{that } \int_0^{\infty} \frac{\omega \sin x \omega}{1 + \omega^2} d\omega = \frac{\pi}{2} e^{-x} \quad (x > 0)$$

Solution : Here we use sine integral formula.

We know that Fourier sine integral of the function $f(x)$ is

$$f(x) = \frac{2}{\pi} \int_0^{\infty} \sin \omega x \int_0^{\infty} f(t) \sin \omega t dt d\omega$$

$$\text{Here } f(t) = \frac{\pi}{2} e^{-t}$$

$$= \frac{2}{\pi} \int_0^{\infty} \sin \omega x \int_0^{\infty} \frac{\pi}{2} e^{-t} \sin \omega t dt d\omega$$

$$= \int_0^{\infty} \sin \omega x \int_0^{\infty} e^{-t} \sin \omega t dt d\omega$$

$$= \int_0^{\infty} \sin \omega x \frac{\omega}{\omega^2 + 1} d\omega$$

$$= \int_0^{\infty} \frac{\omega \sin x \omega}{1 + \omega^2} d\omega$$

Formula

$$\int_0^{\infty} e^{-ax} \sin bx dx = \frac{b}{a^2 + b^2}$$

$$\int_0^{\infty} \frac{\omega \sin x \omega}{1 + \omega^2} d\omega = f(x) = \frac{\pi}{2} e^{-x} \quad (x > 0)$$

Example 2.1.b(4) : Using Fourier integral formula, prove that

$$e^{-ax} - e^{-bx} = \frac{2(b^2 - a^2)}{\pi} \int_0^{\infty} \frac{\lambda \sin \lambda x}{(\lambda^2 + a^2)(\lambda^2 + b^2)} d\lambda, \quad a, b > 0$$

Solution : Here we use the Fourier sine integral formula

$$f(x) = \frac{2}{\pi} \int_0^{\infty} \sin \lambda x \int_0^{\infty} f(t) \sin \lambda t dt d\lambda \quad \dots (1)$$

Here $f(t) = e^{-at} - e^{-bt}$

$$f(x) = \frac{2}{\pi} \int_0^{\infty} \sin \lambda x \left\{ \int_0^{\infty} (e^{-at} - e^{-bt}) \sin \lambda t \, dt \right\} d\lambda$$

$$= \frac{2}{\pi} \int_0^{\infty} \sin \lambda x \left[\int_0^{\infty} e^{-at} \sin \lambda t \, dt - \int_0^{\infty} e^{-bt} \sin \lambda t \, dt \right] d\lambda$$

[Since formula :

$$\int_0^{\infty} e^{-ax} \cos bx \, dx = \frac{a}{a^2 + b^2};$$

$$\int_0^{\infty} e^{-ax} \sin bx \, dx = \frac{b}{a^2 + b^2}$$

Here $a = a, b = \lambda$]

$$= \frac{2}{\pi} \int_0^{\infty} \sin \lambda x \left[\left(\frac{\lambda}{\lambda^2 + a^2} \right) - \left(\frac{\lambda}{\lambda^2 + b^2} \right) \right] d\lambda$$

$$= \frac{2}{\pi} \int_0^{\infty} \frac{\lambda \sin \lambda x [\lambda^2 + b^2 - \lambda^2 - a^2]}{(\lambda^2 + a^2)(\lambda^2 + b^2)} d\lambda$$

$$= \frac{2}{\pi} \int_0^{\infty} \frac{\lambda \sin \lambda x (b^2 - a^2)}{(\lambda^2 + a^2)(\lambda^2 + b^2)} d\lambda$$

$$= \frac{2(b^2 - a^2)}{\pi} \int_0^{\infty} \frac{\lambda \sin \lambda x}{(\lambda^2 + a^2)(\lambda^2 + b^2)} d\lambda$$

Example 2.1.b(5) : Using Fourier integral formula, show that

$$e^{-x} \cos x = \frac{2}{\pi} \int_0^{\pi} \frac{\lambda^2 + 2}{\lambda^4 + 4} \cos \lambda x \, d\lambda$$

Solution : Here we use the Fourier cosine integral formula.

$$f(x) = \frac{2}{\pi} \int_0^{\infty} \cos \lambda x \int_0^{\infty} f(t) \cos \lambda t \, dt \, d\lambda \quad \dots (1)$$

Here $f(t) = e^{-t} \cos t$

$$\begin{aligned}
 f(x) &= \frac{2}{\pi} \int_0^{\infty} \cos \lambda x \int_0^{\infty} e^{-t} \cos t \cos \lambda t dt d\lambda \\
 &= \frac{2}{\pi} \int_0^{\infty} \cos \lambda x \int_0^{\infty} e^{-t} \cos \lambda t \cos t dt d\lambda \\
 &= \frac{2}{\pi} \int_0^{\infty} \cos \lambda x \int_0^{\infty} e^{-t} \left[\frac{\cos (\lambda + 1)t + \cos (\lambda - 1)t}{2} \right] dt d\lambda \\
 &= \frac{2}{\pi} \left(\frac{1}{2} \right) \int_0^{\infty} \cos \lambda x \left[\int_0^{\infty} e^{-t} \cos (\lambda + 1)t dt + \int_0^{\infty} e^{-t} \cos (\lambda - 1)t dt \right] d\lambda \\
 &= \frac{1}{\pi} \int_0^{\infty} \cos \lambda x \left[\frac{1}{1 + (\lambda + 1)^2} + \frac{1}{1 + (\lambda - 1)^2} \right] d\lambda
 \end{aligned}$$

∴ by formula

$$\int_0^{\infty} e^{-ax} \cos bx dx = \frac{a}{a^2 + b^2};$$

$$\int_0^{\infty} e^{-ax} \sin bx dx = \frac{b}{a^2 + b^2}$$

$$\begin{aligned}
 &= \frac{1}{\pi} \int_0^{\infty} \cos \lambda x \left[\frac{1}{\lambda^2 + 2\lambda + 2} + \frac{1}{\lambda^2 - 2\lambda + 2} \right] d\lambda \\
 &= \frac{1}{\pi} \int_0^{\infty} \cos \lambda x \frac{\lambda^2 - 2\lambda + 2 + \lambda^2 + 2\lambda + 2}{(\lambda^2 + 2\lambda + 2)(\lambda^2 - 2\lambda + 2)} d\lambda \\
 &= \frac{1}{\pi} \int_0^{\infty} \cos \lambda x \frac{2(\lambda^2 + 2)}{\lambda^4 - 2\lambda^3 + 2\lambda^2 + 2\lambda^3 - 4\lambda^2 + 4\lambda + 2\lambda^2 - 4\lambda + 4} d\lambda \\
 &= \frac{1}{\pi} \int_0^{\infty} \cos \lambda x \frac{2(\lambda^2 + 2)}{\lambda^4 + 4} d\lambda \\
 &= \frac{2}{\pi} \int_0^{\infty} \cos \lambda x \left(\frac{\lambda^2 + 2}{\lambda^4 + 4} \right) d\lambda
 \end{aligned}$$

Example 2.1.b(6) : Using Fourier integral show that

$$\int_0^{\infty} \frac{1 - \cos \pi \lambda}{\lambda} \sin x \lambda \, d\lambda = \begin{cases} \pi/2 & \text{in } 0 < x < \pi \\ 0 & \text{in } x > \pi \end{cases}$$

Solution : Here we use Fourier sine integral formula

$$f(x) = \frac{2}{\pi} \int_0^{\infty} \sin \lambda x \int_0^{\infty} f(t) \sin \lambda t \, dt \, d\lambda \quad \dots (1)$$

$$f(t) = \begin{cases} \pi/2 & \text{in } 0 < t < \pi \\ 0 & \text{in } t > \pi \end{cases}$$

$$\therefore (1) \Rightarrow f(x) = \frac{2}{\pi} \int_0^{\infty} \sin \lambda x \int_0^{\pi} \left(\frac{\pi}{2} \sin \lambda t \right) dt \, d\lambda$$

$$= \frac{2}{\pi} \left(\frac{\pi}{2} \right) \int_0^{\infty} \sin \lambda x \int_0^{\pi} \sin \lambda t \, dt \, d\lambda$$

$$= \int_0^{\infty} \sin \lambda x \left[\frac{-\cos \lambda t}{\lambda} \right]_0^{\pi} d\lambda$$

$$= \int_0^{\infty} \sin \lambda x \left[\frac{-\cos \lambda \pi}{\lambda} + \frac{1}{\lambda} \right] d\lambda$$

$$= \int_0^{\infty} \frac{1 - \cos \lambda \pi}{\lambda} \sin \lambda x \, d\lambda$$

$$\therefore \int_0^{\infty} \frac{1 - \cos \lambda \pi}{\lambda} \sin \lambda x \, d\lambda = f(x) = \begin{cases} \pi/2 & \text{in } 0 < x < \pi \\ 0 & \text{in } x > \pi \end{cases}$$

Example 2.1.b(7) : Applying the Fourier sine integral formula to the function

$$f(t) = \begin{cases} \sin t & \text{when } 0 < t \leq \pi \\ 0 & \text{when } t > \pi \end{cases}$$

$$\text{show that } \int_0^{\infty} \frac{\sin \omega t \sin \pi \omega}{1 - \omega^2} \, d\omega = \begin{cases} \frac{\pi}{2} \sin t & \text{if } 0 < t < \pi \\ 0 & \text{if } t > \pi \end{cases}$$

Solution : Here we use Fourier sine integral formula

$$f(x) = \frac{2}{\pi} \int_0^{\infty} \sin \lambda x \int_0^{\infty} f(t) \sin \lambda t dt d\lambda \quad \dots (1)$$

$$f(t) = \sin t, \quad 0 < t \leq \pi$$

$$= 0, \quad t > \pi$$

$$\begin{aligned} f(x) &= \frac{2}{\pi} \int_0^{\infty} \sin \lambda x \int_0^{\pi} \sin t \sin \lambda t dt d\lambda \\ &= \frac{2}{\pi} \int_0^{\infty} \sin \lambda x \int_0^{\pi} \sin \lambda t \sin t dt d\lambda \\ &= \frac{2}{\pi} \int_0^{\infty} \sin \lambda x \int_0^{\pi} \frac{\cos(\lambda - 1)t - \cos(\lambda + 1)t}{2} dt d\lambda \\ &= \frac{1}{\pi} \int_0^{\infty} \sin \lambda x \int_0^{\pi} [\cos(\lambda - 1)t - \cos(\lambda + 1)t] dt d\lambda \\ &= \frac{1}{\pi} \int_0^{\infty} \sin \lambda x \left[\frac{\sin(\lambda - 1)t}{\lambda - 1} - \frac{\sin(\lambda + 1)t}{\lambda + 1} \right]_0^{\pi} d\lambda \\ &= \frac{1}{\pi} \int_0^{\infty} \sin \lambda x \left[\frac{\sin(\lambda - 1)\pi}{\lambda - 1} - \frac{\sin(\lambda + 1)\pi}{\lambda + 1} \right] d\lambda \\ &= \frac{1}{\pi} \int_0^{\infty} \sin \lambda x \left[\frac{\sin \lambda \pi \cos \pi - \cos \lambda \pi \sin \pi}{\lambda - 1} - \frac{\sin \lambda \pi \cos \pi + \cos \lambda \pi \sin \pi}{\lambda + 1} \right] d\lambda \\ &= \frac{1}{\pi} \int_0^{\infty} \sin \lambda x \left[\frac{-\sin \lambda \pi}{\lambda - 1} + \frac{\sin \lambda \pi}{\lambda + 1} \right] d\lambda \\ &= \frac{1}{\pi} \int_0^{\infty} \sin \lambda x \sin \lambda \pi \left[\frac{-1}{\lambda - 1} + \frac{1}{\lambda + 1} \right] d\lambda \\ &= \frac{1}{\pi} \int_0^{\infty} \sin \lambda x \sin \lambda \pi \left[\frac{-\lambda - 1 + \lambda - 1}{\lambda^2 - 1} \right] d\lambda \\ &= \frac{1}{\pi} \int_0^{\infty} \sin \lambda x \sin \lambda \pi \left[\frac{-2}{\lambda^2 - 1} \right] d\lambda \end{aligned}$$

$$f(x) = \frac{2}{\pi} \int_0^{\infty} \frac{\sin \lambda x \sin \lambda \pi}{1 - \lambda^2} d\lambda$$

$$\text{i.e., } \int_0^{\infty} \frac{\sin \lambda x \sin \lambda \pi}{1 - \lambda^2} d\lambda = \frac{\pi}{2} f(x)$$

Changing the dummy variable λ to ω we get

$$\int_0^{\infty} \frac{\sin \omega x \sin \omega \pi}{1 - \omega^2} d\omega = \frac{\pi}{2} f(x)$$

$$= \frac{\pi}{2} \sin x \quad \text{for } 0 < x \leq \pi$$

$$= 0 \quad \text{for } x > \pi$$

2.2 FOURIER TRANSFORM PAIR :

2.2.a. Fourier Transform : [Complex Fourier Transform]

Def :- The complex (or infinite) Fourier Transform

The complex (or infinite) Fourier Transform of $f(x)$ is given by

$$F(s) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{isx} dx \quad \dots (1)$$

Then the function $f(x)$ is the inverse Fourier Transform of $F(s)$ and is given by

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(s) e^{-isx} ds \quad \dots (2)$$

The above (1) & (2) are jointly called Fourier transform pair.

Conversion I :

The Fourier transform pair is

$$\text{If } F(s) = \int_{-\infty}^{\infty} f(x) e^{isx} dx \quad \dots (1)$$

$$\text{then } f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-isx} F(s) ds \quad \dots (2)$$